

# Partial Data-Norm Muon: Orthogonalized Updates in the Geometry of the Layer Inputs

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## Abstract

Muon updates each hidden weight matrix with its orthogonalized momentum. That is steepest descent under the spectral norm, which treats every direction of a layer’s input space alike, yet transformer layer inputs are far from isotropic. We study *data-norm Muon*: steepest descent under  $\|\Delta WC^\alpha\|_{\text{op}}$ , where  $C$  is the running second moment of the layer’s inputs. For momentum  $M$  its update is  $\text{polar}(MC^{-\alpha})C^{-\alpha}$ , rescaled to Muon’s size, and  $\alpha = 0$  recovers Muon.

We train a 77M-parameter GPT on FineWeb for 20 tokens per parameter. With  $\alpha = \frac{1}{4}$ , data-norm Muon lowers the final validation loss relative to Muon at its re-tuned learning rate by 0.0193 nats/token on four fresh seeds across three GPU types (L40S, RTX 6000 Ada, A6000; one-sided  $p < 0.001$ ), and by 0.0188 over all six seeds. Equivalently, it reaches tuned Muon’s final loss with about 9% fewer tokens. The gain persists at twice the tokens ( $-0.0154$ ) and at width 768 with 134M parameters ( $-0.0139$ ,  $p \approx 0.006$ ).

Running SOAP-Muon in the data-norm coordinates stacks:  $-0.0313$  against  $-0.0235$  for SOAP-Muon alone, and at width 768, on the one seed where both were run,  $-0.0241$  against  $-0.0183$ . We arrived at the method by tracing an outlier singular value in Muon’s momentum to the gradient a bias would receive, along each layer’s mean input. Exact Hessian-vector products show that loss curvature along an input direction grows in proportion to its variance. Equalizing curvature cost would therefore need the full data norm ( $\alpha = \frac{1}{2}$ ), which gains nothing; the best exponent,  $\alpha = \frac{1}{4}$ , is the square root of that curvature-matched preconditioner. All evidence is at  $\leq 134\text{M}$  parameters, and the wall-clock gain of our unoptimized code is only  $\approx 1.0\text{--}1.07\times$ .

## 1 Introduction

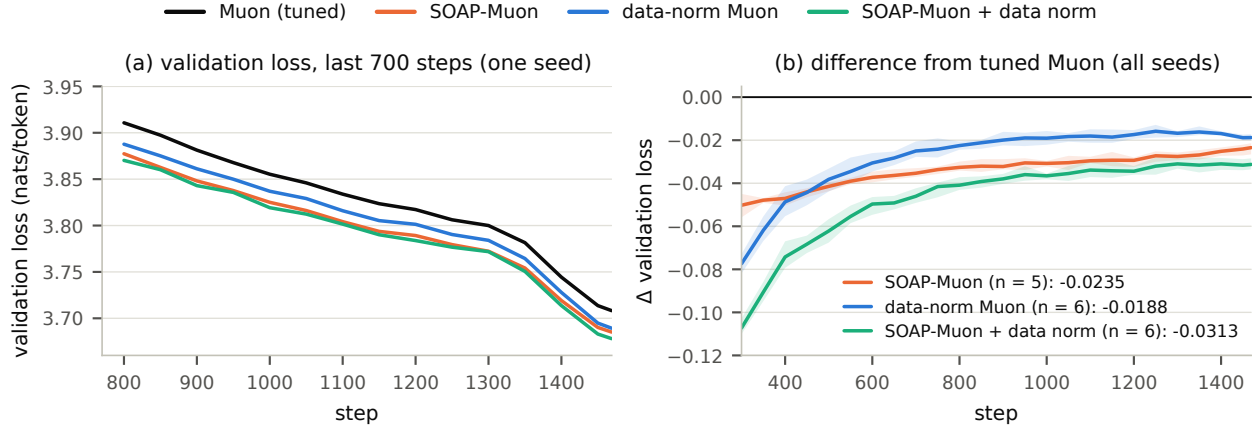
Muon [1] trains the hidden weight matrices of a transformer by replacing each matrix’s momentum  $M$  with its *polar factor*. If  $M = U\Sigma V^\top$  is a singular value decomposition, the polar factor  $\text{polar}(M) = UV^\top$  keeps the singular directions and sets every singular value to one; Muon approximates it with a few Newton–Schulz iterations. Bernstein and Newhouse [2] showed that this is steepest descent under the spectral norm: the step that lowers the loss fastest among steps that move any unit-length input by a bounded amount.

That constraint assumes a layer’s inputs are spread evenly over all directions. They are not. In our models, one shared mean direction carries 5–44% of a layer’s input energy, and even after removing it the top input direction has 57–591 times the median variance (Section 2). A step that moves every input direction equally therefore moves the heavily used directions, and every output that depends on them, much more than the rest.

We ask what happens if Muon measures its step in the geometry of the layer’s actual inputs. The answer is a small change: multiply the momentum by a power of the input second moment before orthogonalizing, and again after. We call it *data-norm Muon*.

### Main findings.

- **It beats a tuned baseline.** Against Muon at its best learning rate, data-norm Muon lowers the final validation loss by 0.0193 nats/token on four fresh seeds and three GPU types (Figure 1). That gap is about 20 times the seed-to-seed noise, and data-norm Muon reaches tuned Muon’s final loss with about 9% fewer tokens.
- **It holds up.** Without re-tuning, about 82% of the gain remains at twice the tokens and 72% at 1.5 times the width (134M parameters).



**Figure 1:** Data-norm Muon reaches a lower validation loss than Muon with a re-tuned learning rate throughout training, and adding SOAP-Muon’s normalization lowers it further (the paper’s model: no biases, RMSNorm, QK-norm; 77M parameters, 20 tokens per parameter). (a) Last 700 steps for one fresh seed; the learning rate decays over the last 10%. (b) Difference from tuned Muon, paired by seed and GPU type: mean over seeds (line) and range (band).

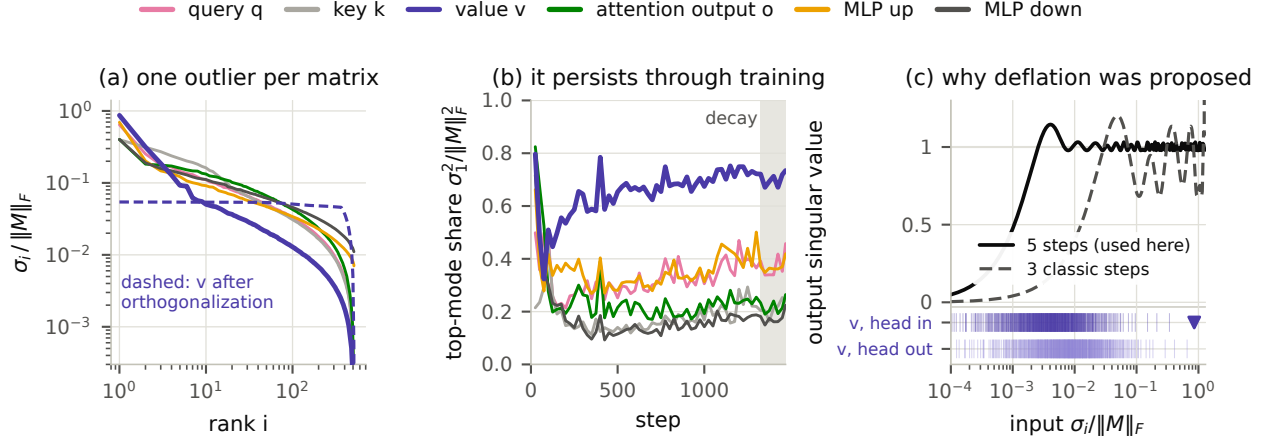
- **It stacks.** Adding SOAP-Muon’s normalization in the data-norm coordinates gives the best method we tested ( $-0.0313$ ).
- **It has limits.** The exponent  $\alpha = \frac{1}{4}$  was chosen by a sweep. It gives the square root of the curvature-matched preconditioner, which works best for reasons we have not tested. All runs have at most 134M parameters, and the wall-clock gain is modest.

## 2 Background: the outlier in Muon’s momentum

This section tells how we arrived at the method. We start from an outlier in the momentum’s spectrum (Figure 2), follow a numerical fix proposed for it, and then trace the outlier to its source (Figure 3). It turns out to be a symptom of anisotropic layer inputs. All measurements here are on the paper’s model (no biases, RMSNorm, QK-norm) at width 512, trained with Muon at learning rate 0.01.

**The observation.** Every body matrix’s momentum has one dominant singular value (Figure 2a). For the value projection  $v$  of block 4, the top singular value holds about three-quarters of the squared norm (0.75 at step 1300), and the rest of the spectrum falls off smoothly behind it. Across the 24 tracked matrices, the top mode’s late-training share  $\rho_1 = \sigma_1^2 / \|M\|_F^2$  is 0.13–0.74. It is there from the first steps and persists through training, including the final decay, and it is largest for  $v$ , at about 0.7 (Figure 2b). Muon at its tuned learning rate shows the same pattern (0.06–0.68). In the depth study that started this work, which used an earlier architecture with biases, the top mode held 0.10–0.87 of the energy. Orthogonalization flattens the whole spectrum (dashed line in panel a), so the outlier’s direction gets exactly the same step size as every other direction.

**The original plan: deflation.** Sun, Kang and Yang [15] argue that such an outlier dominates the normalization before the Newton–Schulz iteration. The iteration divides  $M$  by  $\|M\|_F$  and then pushes every singular value through a fixed polynomial (Figure 2c). When the outlier holds most of the norm, the rest of the spectrum enters at tiny values. The median singular value of  $v$ ’s momentum enters at 0.0038 of the norm, where three classic steps output only 0.16 instead of 1. With the outlier removed from the norm, the input doubles to 0.0074 and the output to 0.30. Deflation does this: it removes the top pair, orthogonalizes the rest and restores the pair. We replicated it as they ran it, with three iterations (one run each). The loss improved from 3.71941 to 3.71144 ( $-0.0080$ ), “consistent but small” by our predeclared reading, and most of the effect came early ( $-0.078$  at step 50).



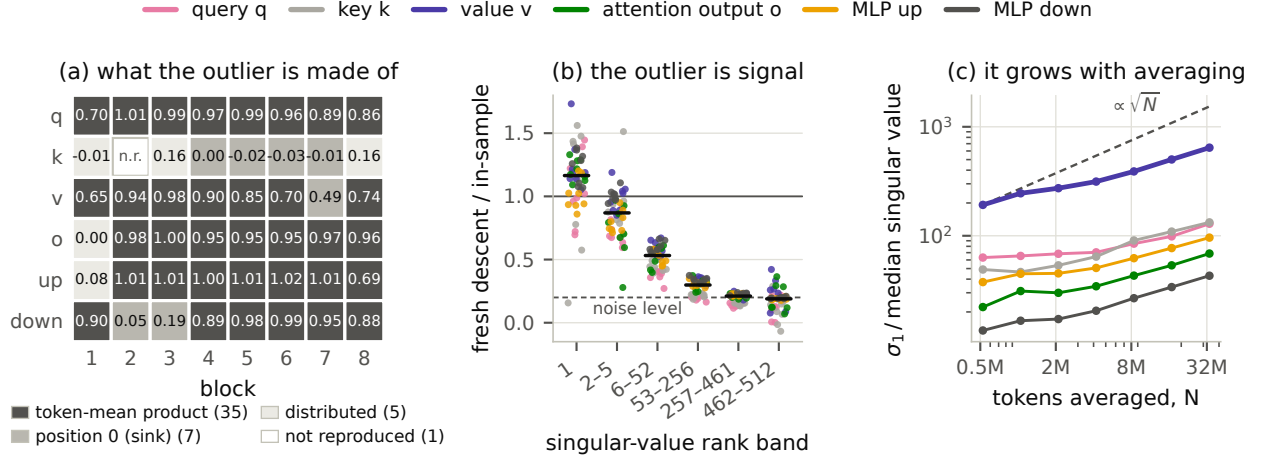
**Figure 2:** Muon’s momentum has one outlier singular value, and it persists through training (the paper’s model: no biases, RMSNorm, QK-norm; Muon at learning rate 0.01). (a) Normalized singular values of the momentum  $M$  for the six matrices of block 4 at step 1300. The dashed line is Muon’s update for  $v$ : orthogonalization makes it flat. (b) The top mode’s share of the momentum’s squared norm (median over blocks 2, 4, 6 and 8); the shaded band is the final learning-rate decay. (c) Newton–Schulz applies a fixed polynomial to each singular value  $\sigma_i / \|M\|_F$ : three classic steps (dashed) or the five steps used here (solid). Ticks show where the singular values of  $v$ ’s final momentum (block 4) enter, with the outlier (triangle) in the norm and with it removed.

This is a numerical fix, not a geometric one. The five steps we use lift everything above about 0.003 of the norm to roughly one (solid line in panel c). They bring 83% of  $v$ ’s singular values above one half, against 16% for three classic steps. On the final momentum they keep 1.00 of an exact polar factor’s first-order descent for every matrix kind. Three classic steps keep 0.71 ( $v$ ) to 0.91 (MLP down), and deflation raises  $v$  to 0.82. Measured on fresh gradients, the five-step update delivers 0.994–1.009 of an exact polar factor’s descent across all 48 matrices, and an exact polar factor later gained nothing against tuned Muon. So the questions became where the outlier comes from, and whether a full-size step along it is right.

**Where the outlier comes from.** We recomputed gradients on never-trained tokens at the final checkpoint and asked what the saved top singular pair ( $u_1, v_1$ ) is made of (Figure 3).

- *It is the gradient a bias would receive* (panel a). The fresh gradient reproduces 47 of the 48 saved outliers. For 35 of them, most of the outlier comes from the *token-mean product*  $\bar{e} \bar{x}^\top$ : the mean output-gradient  $\bar{e}$  times the mean input  $\bar{x}$ . The share is near 1 for most  $q$ ,  $o$ , MLP-up and MLP-down matrices. At the last block it is 0.96 for  $o$ , 0.88 for MLP down and 0.74 for  $v$ , where the first token (an attention sink) accounts for 0.35. In the earlier architecture with biases, the corresponding shares were 0.92–0.98 and 0.75–0.81.
- *The exceptions.* All eight key projections are exceptions: four are dominated by position 0, three by no single source, and one is not reproduced. So are two early MLP-down matrices and one  $v$  dominated by position 0, and two matrices of the first block.
- *The outlier is signal; the bulk is mostly noise* (panel b). Along the top singular pair, fresh gradients reproduce the momentum’s descent in full (median ratio 1.16). Along the bulk they reproduce only 0.2–0.3 of it, and only about three modes per matrix rise above the gradient-noise level.
- *It survives averaging* (panel c). From 0.5M to 33.5M tokens, the top singular value of the fresh mean gradient grows 1.9–4.9 times relative to the median one (across the 48 matrices), while the bulk averages away. The growth falls short of the  $\sqrt{N}$  of a component identical in every batch.
- *Its input side is fixed; its output side drifts.* For the 35 token-mean outliers, the input vector is the mean-input direction (median  $|\cos|$  0.97, range 0.75–1.00). The output vector matches the fresh mean gradient’s only loosely ( $|\cos|$  0.31–0.56). At the last block the mean input concentrates on a few residual-stream channels (participation ratio 4–22, against 140–190 in the middle blocks).

In a network without biases,  $W\bar{x}$  acts as an implicit bias. The mean input carries 5–44% of a layer’s input energy in Muon-trained networks (up to 74% under SOAP-Muon), and the loss is stiff along it. Curvature



**Figure 3:** The outlier is the gradient a bias would receive (the paper’s model: no biases, RMSNorm, QK-norm; fresh gradients from 33.5M never-trained tokens at the final checkpoint of Muon at learning rate 0.01). (a) Share of each matrix’s outlier, its projection  $u_1^\top \bar{G}v_1$  on the fresh mean gradient, explained by the token-mean product  $\bar{e}\bar{x}^\top$ . Shading gives the reading: at least 0.5 from the token-mean product, at least 0.5 from position 0 (the attention sink), neither, or not reproduced by the fresh gradient. Shares can slightly exceed 1 when other terms cancel. (b) Fresh-gradient descent along bands of the momentum’s singular pairs, relative to the momentum’s own (1 = fully reproduced). Points are the 48 matrices, bars are medians, and the dashed line is the predeclared noise level. (c) Top-to-median singular value ratio of the mean fresh gradient of block 4, as more tokens are averaged. A component identical in every batch would grow as  $\sqrt{N}$  (dashed) against a bulk that averages away.

along the  $\bar{x}$  input column is roughly 7–276 times that along random columns, an order-of-magnitude estimate (Appendix B). Muon’s unit-size step along the outlier therefore moves every token’s output together, along a stiff direction.

**What followed.** Treating the outlier directly did not beat tuned Muon (Section 7). Dropping it from the update hurt (by 0.0094 against Muon at the same learning rate). Shrinking the update along  $\bar{x}$  looked slightly better than Muon at learning rate 0.01, but no better than Muon at its re-tuned 0.007. A probe showed that the dropped variant’s deficit is not a missing bias: refitting the implicit biases gained  $\leq 0.0002$ .

The lesson was that the outlier is a symptom. The mean input is only the most extreme direction of a strongly anisotropic input distribution. After centering, the top-to-median eigenvalue ratio is 57–591 and the effective rank is 48–238 of 512. A second clue came from SOAP-Muon, the strongest Muon variant we know: about 80% of its gain comes from input-side preconditioning (Section 6). Both point at the whole input covariance. Hence the design: measure the step, partially, in the norm of the layer’s actual inputs. That norm includes the mean direction but treats all anisotropy consistently.

### 3 Method

The method changes only the norm in which Muon takes its step. A body linear layer computes  $y = Wx$  with  $W \in \mathbb{R}^{m \times n}$  ( $m$  outputs,  $n$  inputs).  $G$  is the minibatch gradient and  $M \leftarrow \mu M + G$  the momentum ( $\mu = 0.95$ ).  $C = \mathbb{E}[xx^\top]$  is the second moment of the layer’s inputs, with eigenvalues  $\lambda_j$  and unit eigenvectors  $q_j$ . Muon’s step is proportional to  $\text{polar}(M)$ , times the shape factor  $\sqrt{\max(1, m/n)}$ .

**The partial data norm.** For  $\alpha \geq 0$  we measure a change  $\Delta W$  by  $\|\Delta W C^\alpha\|_{\text{op}}$ , the largest singular value of  $\Delta W C^\alpha$ . The steepest-descent step maximizes  $\langle M, \Delta W \rangle$  subject to  $\|\Delta W C^\alpha\|_{\text{op}} \leq 1$ . With  $Z = \Delta W C^\alpha$  the objective is  $\langle M C^{-\alpha}, Z \rangle$ , maximized by  $Z = \text{polar}(M C^{-\alpha})$ , so

$$\Delta W^* = \text{polar}(M C^{-\alpha}) C^{-\alpha}. \quad (1)$$

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**Algorithm 1** Data-norm Muon: one optimizer step for a body matrix  $W \in \mathbb{R}^{m \times n}$ .  $\text{NS}(\cdot)$  is Muon’s Newton–Schulz approximation to  $\text{polar}(\cdot)$ .

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**Require:** learning rate  $\eta$ , momentum  $\mu = 0.95$ , weight decay  $\lambda_{\text{wd}} = 0.01$ ,  $\alpha = \frac{1}{4}$ , damping  $\epsilon = 10^{-3}$

*In every training forward pass:*  $\hat{C} \leftarrow 0.998 \hat{C} + 0.002 \cdot \text{mean}_{t \in \{1, 33, 65, \dots\}} x_t x_t^\top$  (bias-corrected)

- 1:  $M \leftarrow \mu M + G$
- 2: **if** first step, or 10 steps since the last refresh **then**
- 3:  $Q \Lambda Q^\top \leftarrow \text{eigh}(\hat{C})$  in double precision;  $R \leftarrow Q (\Lambda / \text{mean}(\Lambda) + \epsilon I)^{-\alpha} Q^\top$
- 4: **end if**
- 5:  $D \leftarrow \text{NS}(MR) R$   $\triangleright$  with SOAP-Muon:  $D \leftarrow \text{NS}(\text{SOAP}(MR; \text{statistics of } GR)) R$
- 6:  $D \leftarrow D \cdot \sqrt{\min(m, n) / \|D\|_F} \cdot \sqrt{\max(1, m/n)}$   $\triangleright$  Muon’s size and shape factor
- 7:  $W \leftarrow (1 - \eta \lambda_{\text{wd}}) W - \eta D$

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$\alpha = 0$  is Muon.  $\alpha = \frac{1}{2}$  is the “full data norm”, which measures output changes on the input distribution itself ( $\mathbb{E} \|\Delta W x\|^2 = \|\Delta W C^{1/2}\|_F^2$ ). We use the partial value  $\alpha = \frac{1}{4}$ . If  $C$  were isotropic apart from its mean,  $C^{-\alpha}$  would shrink only the mean direction; that is the “mean whitening” of Section 2.

**Curvature and the exponent.** Suppose the loss curvature along a rank-one change  $u q_j^\top$  (unit output direction  $u$ ) grows with input variance as  $h_j \propto \lambda_j^\gamma$ , and treat curvature as separable across input directions. A step  $\eta \Delta W$  then costs about  $\frac{\eta^2}{2} \sum_j h_j \|\Delta W q_j\|^2$ . For square and tall matrices ( $m \geq n$ : here  $q, k, v, o$  and MLP-up), a polar factor  $P$  has orthonormal columns, so  $\|P q\| = 1$  for every unit input direction. Muon therefore moves every input direction equally, and its cost per direction grows as  $\lambda_j^\gamma$ : the few high-variance directions, led by the mean input, dominate the cost and cap the learning rate. Equation (1) instead moves  $q_j$  by  $\lambda_j^{-\alpha}$ , for a cost  $\propto \lambda_j^{\gamma-2\alpha}$  that is equal across directions when  $\alpha = \gamma/2$ . We measure  $\gamma \approx 1$  (Section 6), as a Gauss–Newton picture predicts, so full curvature matching is the full data norm,  $\alpha = \frac{1}{2}$ . That gains nothing in training. The best exponent,  $\alpha = \frac{1}{4}$ , gives its square root:  $C^{-1/4} = (C^{-1/2})^{1/2}$ . The argument ignores gradient noise and couplings between directions, and it does not cover the wide MLP down-projection, whose polar factor has orthonormal rows.

**Implementation** (Algorithm 1). Each layer keeps a running average of  $x x^\top$  over every 32nd position of each training sequence (skipping position 0), so no extra forward passes are needed. Each GPU samples 8,192 tokens per step, and the average spans about 16 steps. Every 10 steps one eigendecomposition gives  $R = (C/\bar{\lambda} + 10^{-3} I)^{-\alpha}$ , with  $\bar{\lambda}$  the mean eigenvalue. The update is then rescaled to Muon’s Frobenius norm and shape factor, so only its direction differs from Muon’s. Because it shifts step size toward low-variance directions, it changes the outputs less than Muon at the same norm. It therefore gets its own learning rate (0.01, against Muon’s 0.007), and Section 6 checks that its gain is not just a smaller effective step.

**SOAP-Muon, alone and with the data norm.** Our SOAP-Muon is the preconditioner of the modded-nanogpt optimization-track records [7, 6]. It rotates the momentum into the eigenbases of running averages of  $GG^\top$  and  $G^\top G$ , divides each entry by the root of a running average of its square (close to taking signs on a heavy momentum), rotates back, restores the norm and orthogonalizes. “SOAP-Muon + data norm” runs it on  $MR$  with statistics from  $GR$ , then multiplies by  $R$  and rescales as above.

## 4 Experimental setup

**Model, data and optimizers.** The model is a GPT-style decoder with 8 layers and sequence length 512, at width 512 ( $\approx 77\text{M}$  parameters, including untied embeddings and output head) or width 768 ( $\approx 134\text{M}$ ). It has no biases, RMSNorm with gains, normalized queries and keys, learned positions and a GELU MLP. We train on GPT-2-tokenized FineWeb [18] for 20 tokens per parameter with  $2^{20}$  tokens per step: 1469 steps at width 512 and 2562 at width 768 (the longer runs double the tokens). The learning rate warms up for 50 steps, stays constant, and decays linearly to zero over the last 10% of tokens. Muon trains the six body matrices of every block, with plain momentum 0.95, weight decay 0.01 and five Newton–Schulz iterations

**Table 1:** Final validation loss minus tuned Muon’s, paired by seed, GPU type, width and horizon (nats per token; negative is better). Learning rates: data-norm Muon ( $\alpha = \frac{1}{4}$ ) and SOAP-Muon + data norm 0.01, SOAP-Muon 0.007. “Fresh” excludes the two selection seeds;  $t$  is the paired  $t$  statistic and  $p$  is one-sided. Per-seed values are in Appendix Table 2.

| Method                | Setting                     | Seeds (GPU types) | Mean $\Delta$ | SD     | $t$ (df), $p$          |
|-----------------------|-----------------------------|-------------------|---------------|--------|------------------------|
| Data-norm Muon        | width 512, fresh seeds      | 4 (3)             | −0.0193       | 0.0007 | −56.9 (3), $< 10^{-3}$ |
|                       | width 512, all seeds        | 6 (3)             | −0.0188       | 0.0010 | −43.8 (5), $< 10^{-3}$ |
|                       | width 512, $2\times$ tokens | 2 (2)             | −0.0154       | 0.0014 | —                      |
|                       | width 768, fresh seeds      | 3 (3)             | −0.0139       | 0.0026 | −9.1 (2), 0.006        |
|                       | width 768, all seeds        | 4 (3)             | −0.0140       | 0.0022 | −13.0 (3), $< 10^{-3}$ |
| SOAP-Muon             | width 512                   | 5 (2)             | −0.0235       | 0.0022 | −24.3 (4), $< 10^{-3}$ |
|                       | width 768                   | 1 (1)             | −0.0183       | —      | —                      |
| SOAP-Muon + data norm | width 512, fresh seeds      | 4 (3)             | −0.0316       | 0.0017 | −37.0 (3), $< 10^{-3}$ |
|                       | width 512, all seeds        | 6 (3)             | −0.0313       | 0.0020 | −39.1 (5), $< 10^{-3}$ |
|                       | width 768                   | 2 (2)             | −0.0256       | 0.0020 | —                      |

with per-step coefficients [3]. AdamW (learning rate 0.002) trains embeddings, head and norm gains. Each run uses four GPUs of one type.

**Comparisons and statistics.** The outcome is the final validation loss on  $2^{20}$  held-out tokens, in nats per token. Every comparison pairs two runs with the same seed and GPU type; seeds change only the initialization, and all runs see the same data in the same order. For a fixed recipe the seed-to-seed standard deviation (SD) of the final loss is about 0.001 (0.0006–0.0018 over four seeds on one GPU type), and the same seed on two GPU types can differ by 0.0026. The exponent and learning rates were chosen on two selection seeds; claims rest on fresh seeds. Our claim rule, fixed in advance, requires each method at its bracketed best learning rate, at least three paired fresh seeds on at least two GPU types, every paired difference negative, a mean difference of at most  $-0.005$ , and a one-sided paired  $t$ -test with  $p < 0.05$ . Muon’s learning rate was re-tuned during the study. Its best is 0.007: 0.005 is worse, and the recipe’s default of 0.01 is worse by 0.006–0.009 on all four same-seed pairs (Appendix Table 3). “Tuned Muon” below means Muon at 0.007.

## 5 Results: does it work?

### Key findings.

- Data-norm Muon beats tuned Muon on every seed and GPU type we ran and meets the predeclared claim rule.
- The gain shrinks only modestly at twice the tokens and at width 768.
- Adding SOAP-Muon’s normalization helps further, at both widths.
- The gain is worth about 9% of training tokens, but only 0–7% of wall-clock time with our code.

**The main comparison** (Table 1; every paired difference is plotted in Appendix Figure 6). Data-norm Muon beats tuned Muon on each of the four fresh seeds, by 0.0186–0.0200, on L40S, RTX 6000 Ada and A6000 GPUs. The mean, 0.0193 (SD 0.0007,  $p < 0.001$ ), meets every clause of the claim rule; with the two selection seeds it is 0.0188. The gap opens early and does not close during the final learning-rate decay (Figure 1b). Data-norm Muon’s own learning rate is bracketed: 0.007 and 0.014 are both worse. It recovers about 80% of SOAP-Muon’s gain over tuned Muon, but SOAP-Muon alone is still better, by 0.003–0.007 on each of the five seeds where both ran.

**Longer training and a wider model.** We kept  $\alpha$ , the learning rates and all other settings fixed. With twice the tokens, data-norm Muon gains 0.0164 and 0.0145 on two seeds, about 82% of its standard-budget gain. At width 768, re-bracketing both methods’ learning rates found the same best values, and three fresh seeds gain 0.0118, 0.0169 and 0.0130: a mean of 0.0139 ( $p \approx 0.006$ ), about 72% of the width-512 gain.

Our criteria, fixed in advance, asked every fresh seed to gain at least 0.012. One missed by 0.0002, which triggered the rule’s “add a seed” step (the added seed gained 0.0130). By the letter of the rule the result is “ambiguous”. It is far from “weakened”, which required a mean gain under 0.009 or any seed at or above zero.

**Stacking with SOAP-Muon.** SOAP-Muon + data norm gains 0.0313 over tuned Muon at width 512 across six seeds and three GPU types (0.0316 on the four fresh seeds). With its learning rate bracketed, it also meets the claim rule. On every seed where both ran, it beats SOAP-Muon alone by 0.006–0.009 and data-norm Muon by 0.011–0.015. At width 768, with the width-512 learning rates, it gains 0.0241 and 0.0270 on two seeds, against 0.0183 for SOAP-Muon alone on the one seed we completed. The stacking therefore also holds at the larger width, on that seed.

**Tokens and wall-clock time.** We trained data-norm Muon on 70% and 85% of the token budget, each with its own learning-rate decay. Interpolating, it matches tuned Muon’s final loss at about 91% of the tokens, a  $\approx 1.10\times$  token efficiency. Tuned Muon improves by 0.107 per doubling of tokens; at that rate the gains correspond to  $\approx 1.13\times$  (data-norm Muon),  $\approx 1.16\times$  (SOAP-Muon) and  $\approx 1.22\times$  (SOAP-Muon + data norm). Our unoptimized code adds 6%, 9% and 15% to the median step time (Appendix Table 4), which leaves wall-clock gains of only  $\approx 1.0$ – $1.07\times$ . Data-norm Muon’s overhead is the covariance average and the eigendecomposition every 10 steps, both easy to reduce.

## 6 Why it works: a few diagnostics

### Key findings.

- The gain comes from the direction of the step, not from a hidden change in step size.
- Loss curvature grows in proportion to input variance, so full curvature matching would be  $\alpha = \frac{1}{2}$ . The best exponent,  $\frac{1}{4}$ , gives the square root of that preconditioner.
- SOAP-Muon’s gain is mostly on the input side, yet it stacks with the data norm: the two do different things.

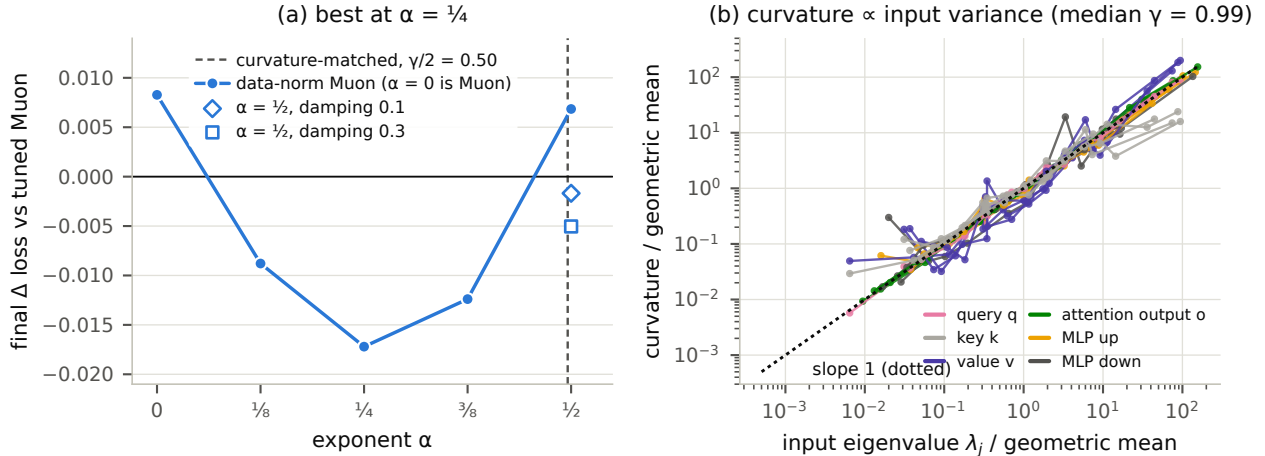
Each diagnostic is a single run on one selection seed at learning rate 0.01 unless noted. Appendix Table 5 lists every ablation against both tuned Muon and Muon at 0.01.

**Direction, not step size.** Data-norm Muon moves the outputs less than Muon at the same norm, so part of its effect could be a hidden per-layer learning-rate cut. We therefore kept Muon’s direction and rescaled it, per layer and per step, to data-norm Muon’s output energy  $\mathbb{E}\|\Delta Wx\|^2$ . This magnitude-matched Muon ends level with tuned Muon (−0.0017), so the gain is geometric. Two controls are also null: switching on the statistics layers alone gives +0.0007, and an exact polar factor is 0.0052 behind tuned Muon.

**The exponent** (Figure 4a). Against tuned Muon,  $\alpha = \frac{1}{8}, \frac{1}{4}, \frac{3}{8}$  and  $\frac{1}{2}$  give −0.0088, −0.0172, −0.0124 and +0.0068: the full data norm gains nothing. Both halves of the sandwich matter. Whitening only inside the polar factor (exponent  $\frac{1}{2}$ ) gives −0.0064, and only outside ( $\frac{1}{4}$ ) gives −0.0075. Centering  $C$  halves the gain (−0.0089), so the mean direction matters as part of the full anisotropy.

**What the curvature says.** We measured the curvature exactly, with Hessian-vector products along  $uq_j^\top$ , for 12 input eigendirections of each of 24 matrices of tuned Muon (Figure 4b).

- *Curvature is proportional to input variance.* The median slope  $\gamma$  of log curvature against log variance is 0.99 at width 512 (by kind:  $q$  1.00,  $k$  0.78,  $v$  1.02,  $o$  1.02, MLP up 0.92, MLP down 0.96) and 0.99 at width 768, as a Gauss–Newton picture predicts. The key projection is flatter at both widths (0.78 and 0.74), plausibly because of the QK-norm.
- *So curvature matching does not explain the exponent.* By the equal-cost argument of Section 3, the curvature-matched exponent is  $\alpha = \gamma/2 \approx \frac{1}{2}$ , the full data norm, which gains nothing in training.



**Figure 4:** The best exponent is half the curvature-matched one (the paper’s model: no biases, RMSNorm, QK-norm; width 512). (a) Final loss against the exponent  $\alpha$  (one seed, learning rate 0.01;  $\alpha = 0$  is Muon at that rate). Open markers are the full data norm with damping (Section 7), and the dashed line is the curvature-matched exponent  $\gamma/2$  from panel (b). (b) Exact loss curvature, from Hessian-vector products, along input eigendirections against their variance, for 24 matrices of tuned Muon, each normalized by its geometric means. The dotted line has slope 1: curvature is proportional to variance, except for the flatter key projections.

- *The best preconditioner is its square root.*  $\alpha = \frac{1}{4}$  gives  $C^{-1/4}$ , the square root of the curvature-matched  $C^{-1/2}$ : halfway, in log scale, between Muon ( $\alpha = 0$ ) and curvature matching ( $\alpha = \frac{1}{2}$ ).

A plausible but untested reading: if gradient noise also scales with curvature, the square root of the curvature preconditioner is preferable to the full inverse. That is the square root Adagrad [4] and Shampoo [5] use. Full whitening would amplify noise-dominated low-variance directions. We chose  $\alpha = \frac{1}{4}$  by a sweep and transferred it unchanged to width 768. An earlier finite-difference estimate ( $\gamma \approx 0.5$ ) was a float32 resolution artifact (Appendix B).

**What carries SOAP-Muon’s gain.** Keeping only SOAP’s input-side eigenbasis retains 77–83% of its gain over Muon at learning rate 0.01 (four seeds, three GPU types). A basis built from the input second moment instead of gradient statistics keeps 74%. The gain sits on the square and tall matrices; applied to the MLP down-projection alone, SOAP-Muon gains nothing against tuned Muon. On those matrices orthogonalization fixes how far each input direction moves, so SOAP can only re-route inputs to outputs, while the data norm changes the distances. Consistent with two distinct mechanisms, SOAP’s output-side basis alone, run in data-norm coordinates, recovers 79% of the stacking gain.

The two methods also help different tokens. Binned by target-token frequency, SOAP-Muon’s gain leans toward rare targets: 0.032–0.038 for frequencies  $10^{-6}$ – $10^{-4}$ , against 0.011 for the most frequent. That is the pattern expected of Adam-like normalization on heavy-tailed data [16]. The data norm’s gain is broad, 0.016–0.020 even on frequent tokens.

## 7 What did not work

Appendix Table 6 lists every run in this section.

**Treating the outlier directly.** We tried whitening only the mean input, removing the update along it, dropping the outlier, capping its step while raising the rest, and deflation. Against Muon at learning rate 0.01, several gained early and mean whitening ended slightly ahead (−0.0033 over three runs). Against tuned Muon, all ended level or behind, by 0.002–0.018. Their early advantage has the same shape as Muon at 0.02 minus Muon at 0.01, a larger effective step: the lever was Muon’s untuned learning rate, not the outlier. Deflation, run only with three iterations, ended 0.0046 behind tuned five-iteration Muon. Nesterov momentum changed nothing (+0.0001).

**Two refinements of the data norm.** Both were motivated by the earlier, resolution-limited finite-difference curvature estimates.

- *Per-kind exponents.* Setting  $\alpha = \gamma/2$  separately for each matrix kind (0.17–0.36 from those estimates) gave a mean of  $-0.0171$ , against  $-0.0177$  for  $\alpha = \frac{1}{4}$  on the same seeds.
- *Damped full data norm.* A linear-plus-floor fit to the same data suggested  $\alpha = \frac{1}{2}$  with damping of about 0.1 of the mean eigenvalue. It gave only  $-0.0017$  and  $-0.0066$  (and  $-0.0050$  with damping 0.3), against  $-0.0172$  and  $-0.0181$  for  $\alpha = \frac{1}{4}$ .

Together with the undamped full data norm, these runs show that curvature matching, however parametrized, does not produce the best exponent.

**Predictions that failed.** We recorded predictions before each experiment (Appendix Table 8). Four about the final method failed:

- *Magnitude matching* (P16) was to keep under a third of the gain over Muon at 0.01. It kept 39%, but only about 10% against tuned Muon.
- *SOAP’s gain* (P17) was to be mostly input anisotropy, leaving little to stack. Stacking added 0.0112.
- *Per-kind exponents* (P20) were to gain at least 0.002 on both seeds. They gave  $-0.0007$  and  $+0.0019$ .
- *The damped full data norm* (P21) was to come within 0.003 of  $\alpha = \frac{1}{4}$ . It fell short by more than 0.01.

Four earlier mechanism predictions also failed (P7, P8, P9, P13). One of them, that Muon at 0.007 would match Muon at 0.01, is how we found that the baseline needed re-tuning.

## 8 Related work

**Muon and preconditioning.** Bernstein and Newhouse [2] cast Muon [1] as steepest descent under an operator norm; data-norm Muon keeps the orthogonalization and changes the norm. Our recipe follows [3]. K-FAC [8] preconditions with Kronecker factors whose input factor is the activation second moment  $C$ . Shampoo [5] uses Kronecker factors of gradient statistics, and SOAP [6] runs Adam in Shampoo’s eigenbasis. The modded-nanogpt records [7] precondition Muon’s momentum with a SOAP-style step, our SOAP-Muon.

**Activation statistics in Muon.** GO-MUON [9] proposes an activation-covariance sandwich on both sides of the polar factor, with exponent  $\frac{1}{4}$ , at toy scale. It is the closest prior work, and the form  $\text{polar}(MC^{-1/4})C^{-1/4}$  is not new. What appears new here is the input-only, full-matrix version tested at 77–134M parameters against a tuned baseline, the measurement that curvature grows in proportion to input variance (so the best exponent gives the square root of the curvature-matched preconditioner), the stacking with SOAP-Muon, and the path from the momentum outlier. Newton-Muon [10] multiplies the gradient by a damped inverse of the activation second moment before the momentum. PMuon (a modded-nanogpt record) applies second-moment powers on both sides inside the polar factor. IsoMuon [11] uses a diagonal, clamped variant, Mousse [12] a milder whitening, and NorMuon [13] neuron-wise normalization of Muon’s update. Neither Newton-Muon nor PMuon multiplies the orthogonalized update on the right. In one single-seed screen on H100, with only their preconditioners ported and their learning rates untuned, both ended within 0.003 of tuned Muon, against 0.0186 for data-norm Muon; we draw no conclusion from it.

**The outlier, heavy tails and benchmarking.** SAMuon [14] finds that Muon’s rank-one head sits at the edge of stability and limits the learning rate; spectral deflation [15] removes it before orthogonalizing. Kunstner et al. [16] tie Adam’s advantage on language to heavy-tailed class imbalance, which matches SOAP-Muon’s rare-token profile. Wen et al. [17] find that reported optimizer speedups shrink under careful tuning and with scale, which motivates our re-tuned baseline and our caution.

## 9 Limitations and conclusion

**Limitations.** All runs have at most 134M parameters and 8 layers, on one dataset with one batch size. The gain shrinks modestly with width and tokens (to about 72% and 82%), each measured on few seeds,

and the width-768 result misses its strict per-seed bar by 0.0002 on one seed. Most diagnostics use one seed, and the width-768 SOAP-Muon comparison rests on one seed with carried-over learning rates. The exponent  $\alpha = \frac{1}{4}$  was chosen by a sweep and transferred unchanged to width 768; why the square root of the curvature-matched preconditioner works best is untested. The wall-clock gain is only  $\approx 1.0\text{--}1.07\times$  with our current code. We therefore state the result as holding across  $1.5\times$  width and  $2\times$  tokens at up to 134M parameters, not as a scaling law.

**Conclusion.** An outlier in Muon’s momentum led us to the geometry of the layer inputs. Muon’s orthogonalized step treats the input space as isotropic, and on square and tall matrices no preconditioner that ends in orthogonalization can change that. Multiplying by a fractional power of the input second moment,  $\text{polar}(MC^{-1/4})C^{-1/4}$ , reallocates step size across input directions using only forward-pass statistics. It consistently beats a re-tuned Muon across seeds and GPU types, survives twice the tokens and 1.5 times the width without re-tuning, and combines with SOAP-Muon into the best method we tested. Whether the gain persists at larger scale, and why the square root of the curvature-matched preconditioner works best, remain open.

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## A Detailed results

Short labels: PD = data-norm Muon ( $\alpha = \frac{1}{4}$ ), S = SOAP-Muon, S+PD = SOAP-Muon + data norm. *sel.* marks the selection seeds (260925 and 260926 at width 512; 260925, the learning-rate bracket seed, at width 768); *fresh* marks confirmation seeds. RTX = RTX 6000 Ada.

**Table 2:** Per-seed final  $\Delta$  against tuned Muon (learning rate 0.007) on the same seed and GPU type. PD and S+PD use learning rate 0.01, S uses 0.007.

| Seed                                    | GPU   | Role  | Tuned Muon | PD      | S                              | S+PD    |
|-----------------------------------------|-------|-------|------------|---------|--------------------------------|---------|
| <i>Width 512, 1469 steps</i>            |       |       |            |         |                                |         |
| 260924                                  | L40S  | fresh | 3.70427    | −0.0186 | −0.0216                        | −0.0302 |
| 260924                                  | RTX   | fresh | 3.70689    | −0.0198 | (−0.0249 at 0.01) <sup>a</sup> | −0.0336 |
| 260925                                  | A6000 | sel.  | 3.70630    | −0.0172 | −0.0214                        | −0.0284 |
| 260926                                  | A6000 | sel.  | 3.70947    | −0.0181 | −0.0246                        | −0.0327 |
| 260927                                  | A6000 | fresh | 3.70823    | −0.0189 | −0.0231                        | −0.0302 |
| 260928                                  | A6000 | fresh | 3.71039    | −0.0200 | −0.0266                        | −0.0325 |
| <i>Width 512, 2× tokens, 2938 steps</i> |       |       |            |         |                                |         |
| 260924                                  | L40S  | fresh | 3.59730    | −0.0164 | —                              | —       |
| 260925                                  | RTX   | sel.  | 3.59632    | −0.0145 | —                              | —       |
| <i>Width 768, 2562 steps</i>            |       |       |            |         |                                |         |
| 260925                                  | A6000 | sel.  | 3.46974    | −0.0142 | —                              | —       |
| 260924                                  | L40S  | fresh | 3.47212    | −0.0118 | −0.0183                        | −0.0241 |
| 260926                                  | RTX   | fresh | 3.47526    | −0.0169 | stopped <sup>b</sup>           | −0.0270 |
| 260927                                  | A6000 | fresh | 3.46983    | −0.0130 | —                              | —       |

<sup>a</sup> Not in the S mean, which uses learning rate 0.007. <sup>b</sup> Stopped at about step 270 of 2562 at the user’s request; not a result. S+PD minus S on the same seeds: −0.0086, −0.0071, −0.0080, −0.0071, −0.0060 (260924 L40S; 260925–260928), and −0.0058 at width 768. S+PD minus PD: −0.0112 to −0.0145 (six seeds).

**Table 3:** Learning-rate brackets: final validation loss, best per row in bold.

| Method | Width | Seed, GPU     | 0.005   | 0.007          | 0.01           | 0.014   | 0.02    |
|--------|-------|---------------|---------|----------------|----------------|---------|---------|
| Muon   | 512   | 260925, A6000 | 3.71008 | <b>3.70630</b> | 3.71457        |         |         |
| Muon   | 512   | 260926, A6000 |         | <b>3.70947</b> | 3.71505        |         |         |
| Muon   | 512   | 260924, L40S  |         | <b>3.70427</b> | 3.71310        | 3.71900 | 3.72007 |
| Muon   | 512   | 260924, RTX   |         | <b>3.70689</b> | 3.71366        |         |         |
| PD     | 512   | 260925, A6000 |         | 3.69343        | <b>3.68909</b> | 3.69570 |         |
| PD     | 512   | 260926, A6000 |         | 3.69318        | <b>3.69133</b> |         |         |
| PD     | 512   | 260924, L40S  |         | 3.68595        | <b>3.68569</b> |         |         |
| S      | 512   | 260925, A6000 |         | <b>3.68490</b> | 3.68699        |         |         |
| S      | 512   | 260926, A6000 |         | <b>3.68486</b> | 3.68808        |         |         |
| S      | 512   | 260924, L40S  |         | 3.68263        | <b>3.68106</b> | 3.68805 |         |
| S+PD   | 512   | 260925, A6000 |         | 3.67901        | <b>3.67785</b> | 3.68087 |         |
| Muon   | 768   | 260925, A6000 | 3.47205 | <b>3.46974</b> | 3.47610        |         |         |
| PD     | 768   | 260925, A6000 |         | 3.45742        | <b>3.45549</b> | 3.45894 |         |

**Table 4:** Step time on one node of four L40S GPUs (seed 260924; unoptimized code) and estimated wall-clock speedups (token speedup divided by relative step time; the token speedup is measured for PD and extrapolated for S and S+PD).

|                          | Median step<br>(s) | Overhead vs Muon |       | Speedup         |                         |
|--------------------------|--------------------|------------------|-------|-----------------|-------------------------|
|                          |                    | median           | mean* | tokens          | wall clock              |
| Muon (tuned)             | 0.894              | —                | —     | 1               | 1                       |
| PD                       | 0.946              | +6%              | +9%   | 1.10 (measured) | $\approx 1.01$ – $1.04$ |
| S                        | 0.971              | +9%              | +8%   | $\approx 1.16$  | $\approx 1.06$ – $1.07$ |
| S, input-side basis only | 0.930              | +4%              | —     | —               | —                       |
| S+PD                     | 1.028              | +15%             | +19%  | $\approx 1.22$  | $\approx 1.03$ – $1.06$ |

\*Means are sensitive to occasional slow steps, which cluster in the final decay: over all steps they are +11%, +8% and +22%; over steps 51–1322, +9%, +8% and +18%. Token runs at 70% and 85% of the budget (seed 260925, A6000) reached 3.76158 and 3.71833, against 3.68909 for PD and 3.70630 for tuned Muon at the full budget.

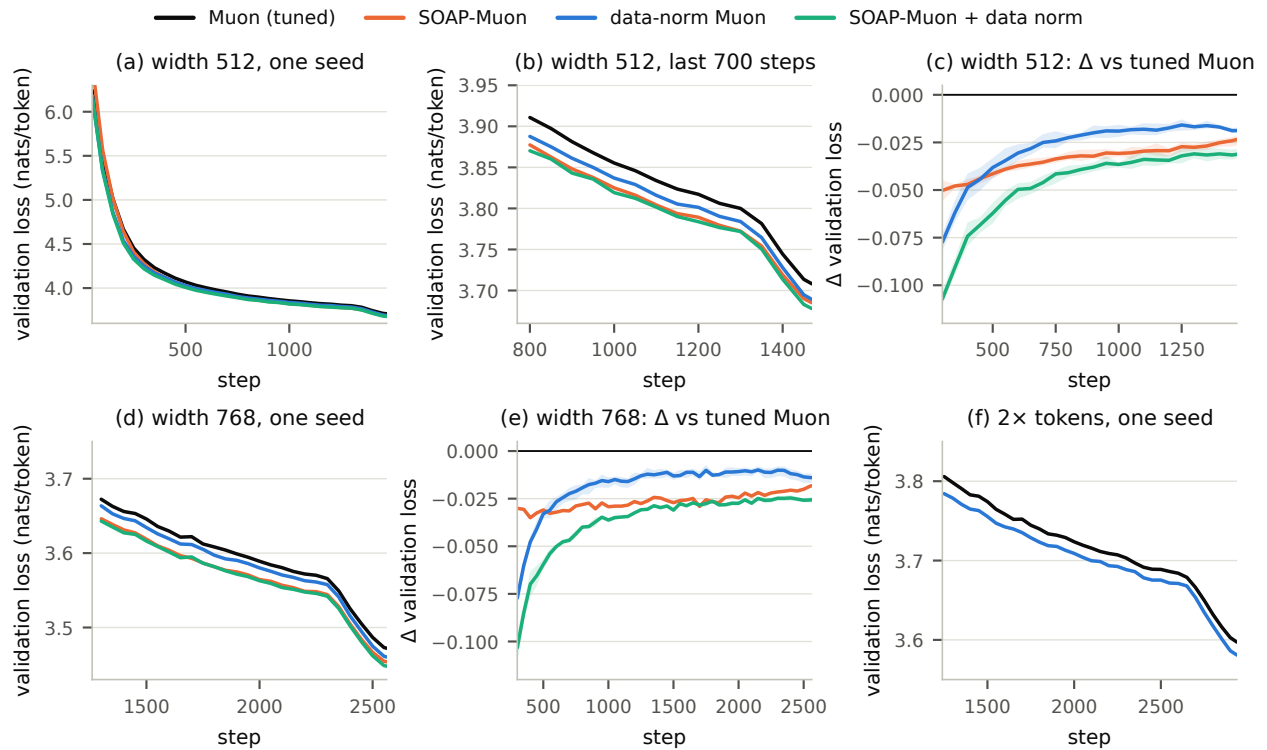
**Table 5:** Ablations and controls: final  $\Delta$  against tuned Muon (learning rate 0.007) and against Muon at 0.01, on the same seed and GPU. Single runs on seed 260925 (A6000) at learning rate 0.01 unless noted. “Share” is the fraction of S’s gain over Muon at 0.01 on the same seed and GPU (the predeclared dissection metric).

| Run                                          | vs tuned Muon  | vs Muon at 0.01 | Share / note     |
|----------------------------------------------|----------------|-----------------|------------------|
| <i>Data-norm variants</i>                    |                |                 |                  |
| PD, $\alpha = 1/4$ (sandwich)                | −0.0172        | −0.0255         |                  |
| $\alpha = 1/8$                               | −0.0088        | −0.0171         |                  |
| $\alpha = 3/8$                               | −0.0124        | −0.0207         |                  |
| $\alpha = 1/2$ (full data norm)              | +0.0068        | −0.0014         |                  |
| Inside only, polar( $MC^{-1/2}$ )            | −0.0064        | −0.0147         |                  |
| Outside only, polar( $M$ ) $C^{-1/4}$        | −0.0075        | −0.0158         |                  |
| Centered covariance                          | −0.0089        | −0.0172         |                  |
| PD + output-row normalization                | −0.0161        | −0.0243         |                  |
| Magnitude-matched Muon                       | −0.0017        | −0.0099         |                  |
| <i>Controls</i>                              |                |                 |                  |
| Statistics layers only (Muon at 0.007)       | +0.0007        | —               |                  |
| Exact polar factor, Muon at 0.01             | +0.0052        | −0.0031         |                  |
| <i>SOAP dissection</i>                       |                |                 |                  |
| S, both sides                                | −0.0193        | −0.0276         | share 1.00       |
| Input-side basis only (4 seeds, 3 GPU types) | −0.0161 (mean) | −0.0234 (mean)  | share 0.77–0.83  |
| with an activation second-moment basis       | −0.0122        | −0.0205         | share 0.74       |
| one denominator per eigendirection           | −0.0062        | −0.0145         | share 0.53       |
| Standard basis (no rotation)                 | +0.0033        | −0.0050         | share 0.18       |
| Gradient, not momentum, second moment        | −0.0190        | −0.0273         | share 0.99       |
| S on MLP-down only                           | +0.0001        | −0.0082         | share 0.30       |
| S on all but MLP-down                        | −0.0173        | −0.0256         | share 0.93       |
| <i>Stacking</i>                              |                |                 |                  |
| S+PD                                         | −0.0284        | −0.0367         |                  |
| Output-side S only, in data-norm coordinates | −0.0261        | −0.0344         | 79% of S+PD − PD |

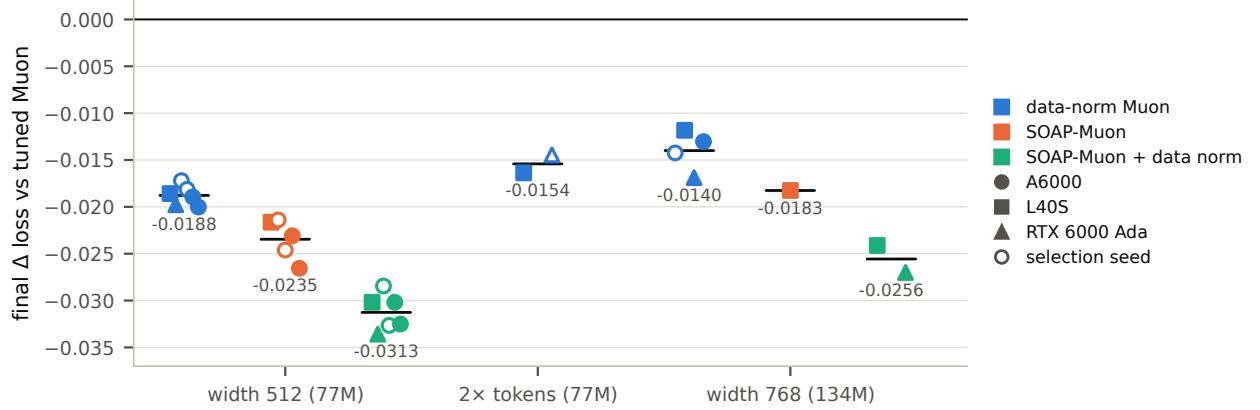
**Table 6:** Runs that did not beat tuned Muon, or did not improve on uniform PD.  $\Delta$  is on the same seed and GPU type.

| Run (learning rate 0.01 unless noted)                                         | Seed, GPU           | vs tuned Muon        | vs Muon, same rate   |
|-------------------------------------------------------------------------------|---------------------|----------------------|----------------------|
| <i>Treating the outlier directly</i>                                          |                     |                      |                      |
| Mean whitening, automatic strength                                            | 3 runs <sup>a</sup> | +0.0047 (mean)       | −0.0033 (mean)       |
| Mean whitening, $\beta = 1/4$                                                 | 260925, A6000       | +0.0022              | −0.0061              |
| No update along $\bar{x}$ ( $\beta = 0$ )                                     | 260924, RTX         | +0.0124              | +0.0056              |
| Drop the tracked outlier from the update                                      | 260924, RTX         | +0.0161              | +0.0094              |
| Cap the outlier: it at 0.01, the rest at 0.02                                 | 260925, A6000       | +0.0176              | +0.0093 <sup>b</sup> |
| Spectral deflation, 3 iterations                                              | 260924, RTX         | +0.0046 <sup>c</sup> | −0.0080 <sup>d</sup> |
| Nesterov momentum                                                             | 260924, RTX         | +0.0068              | +0.0001              |
| <i>Data-norm refinements</i> (uniform PD on the same seeds: −0.0172, −0.0181) |                     |                      |                      |
| Per-kind $\alpha = \gamma/2$ (finite-difference $\gamma$ )                    | 260925/6, A6000     | −0.0179, −0.0162     | —                    |
| Damped full data norm, $\alpha = 1/2$ , damping 0.1                           | 260925/6, A6000     | −0.0017, −0.0066     | —                    |
| Damped full data norm, $\alpha = 1/2$ , damping 0.3                           | 260925, A6000       | −0.0050              | —                    |

<sup>a</sup> Seed 260924 on RTX and on L40S; seed 260925 on A6000. <sup>b</sup> vs Muon at 0.01. <sup>c</sup> vs tuned five-iteration Muon, same seed and GPU. <sup>d</sup> vs plain three-iteration Muon, run on L40S (cross-hardware, single seed); the deflated run reached 3.71144 and the plain one 3.71941.



**Figure 5:** Training curves for all settings (the paper’s model: no biases, RMSNorm, QK-norm). (a,b) Width 512, one fresh seed (260927, A6000). Finals: tuned Muon 3.7082, SOAP-Muon 3.6851, data-norm Muon 3.6893, SOAP-Muon + data norm 3.6780. (d) Width 768, seed 260924 (L40S). Finals: 3.4721, 3.4539, 3.4603 and 3.4480 in the same order. (f) Twice the tokens, seed 260924 (L40S): tuned Muon 3.5973, data-norm Muon 3.5809. (c,e) Paired differences from tuned Muon: mean over seeds (line) and range (band); final means at width 512 are −0.0235 (SOAP-Muon, 5 seeds), −0.0188 (data-norm Muon, 6) and −0.0313 (SOAP-Muon + data norm, 6), and at width 768 −0.0183 (1 seed), −0.0140 (4) and −0.0256 (2).



**Figure 6:** Every paired final difference from tuned Muon (the paper’s model: no biases, RMSNorm, QK-norm). Colour marks the method, marker shape the GPU type, and hollow markers the selection seeds; bars are means. The width-768 SOAP-Muon runs reuse the width-512 learning rates. Curves over training for every setting are in Figure 5.

## B Implementation details and hyperparameters

**Table 7:** Hyperparameters shared by all runs unless stated.

|                     |                                                                                                                                                                                                                            |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Architecture        | depth 8; width 512 (8 heads) or 768 (12 heads); head dimension 64; sequence length 512; vocabulary 50,304 (GPT-2, padded); no biases; RMSNorm with gains ( $\epsilon = 10^{-6}$ ); per-head query/key RMSNorm; GELU 4× MLP |
| Initialization      | $\mathcal{N}(0, 0.02^2)$ ; attention-output and MLP-down scaled by $1/\sqrt{2 \cdot \text{depth}}$                                                                                                                         |
| Batch and budget    | $2^{20}$ tokens per step; 20 tokens per parameter (1469 or 2562 steps); 2× runs: 2938 steps                                                                                                                                |
| Schedule            | 50-step linear warmup, constant, linear decay to 0 over the last 10% of tokens; gradient clipping at norm 1                                                                                                                |
| Muon                | $q, k, v, o$ , MLP-up, MLP-down (48 matrices); plain momentum 0.95; weight decay 0.01; shape factor $\sqrt{\max(1, m/n)}$                                                                                                  |
| Newton–Schulz       | 5 steps with distinct quintic coefficients per step [3]; BF16 products, FP32 normalization                                                                                                                                 |
| AdamW               | learning rate 0.002, $\beta = (0.9, 0.95)$ , $\epsilon = 10^{-8}$ , weight decay 0.01; embeddings, head and gains                                                                                                          |
| Precision, hardware | FP32 parameters and states, BF16 autocast; 4 GPUs per run (A6000, L40S or RTX 6000 Ada)                                                                                                                                    |
| Data norm           | $\alpha = 1/4$ ; damping $10^{-3}$ of the mean eigenvalue; FP64 eigendecomposition every 10 steps                                                                                                                          |
| Input statistics    | every 32nd position from position 1; running-average decay 0.998 per micro-batch forward                                                                                                                                   |
| SOAP                | Gram averages $\beta_2 = 0.9$ with one QR refresh per step; denominator average $\beta_2 = 0.9$ , power $1/2$                                                                                                              |
| Learning rates      | Muon 0.007; PD 0.01; S 0.007; S+PD 0.01                                                                                                                                                                                    |

Each matrix’s orthogonalized update is computed on one owner GPU (round-robin within each shape) and shared with one all-reduce, so every replica applies the same update. The data-norm and SOAP statistics live on that owner GPU and are not checkpointed. The statistics layers use non-persistent buffers, so initial weights and their hashes are unchanged, and they do not change training (Table 5). Before each run, a tiny multi-GPU run audited replica consistency and a five-step full-size run qualified the recipe.

### Measurement notes.

- *Momentum spectra* (Figure 2). From Muon at learning rate 0.01 (seed 260924, L40S), recorded every 25 steps; the late-training shares average steps 1300–1469. The tuned-Muon comparison uses seed 260925 (A6000). The first-order descent kept by three and five Newton–Schulz steps is computed on the final momentum of the same run.
- *Outlier attribution* (Figure 3). 64 batches of  $2^{19}$  never-trained tokens at that run’s final checkpoint. Readings follow the predeclared rule: a reproduced outlier is “token-mean product” or “position 0” when that term explains at least 0.5 of  $u_1^\top \bar{G} v_1$ ; bands count as signal at a ratio  $\geq 0.5$  and as noise below 0.2.
- *Input anisotropy*. 128 validation sequences, final checkpoints of Muon and SOAP-Muon at learning rate 0.01 (seed 260925).

- *Curvature* (Figure 4b). Exact Hessian-vector products (double backpropagation in FP32 with the math attention kernel) along  $u q_j^\top$  for 12 input eigendirections of each of 24 matrices, averaged over 3 random unit output vectors  $u$ , on 64 fresh sequences at tuned Muon’s final weights (seed 260925). At width 512 all 288 curvatures are positive; at width 768, 287 are (median slope 0.985). An earlier finite-difference version quantized its second differences at the float32 spacing of the loss. Most low-variance directions were then unresolved, and its fitted  $\gamma \approx 0.5$  was an artifact; the per-kind exponents of Table 6 used those estimates.
- *Mean-input curvature ratio* (7–276). From a similar probe on the learning-rate-0.01 run, where the random-direction curvatures are only 1–4 float32 rounding units.

## C Predeclared predictions and outcomes

**Table 8:** Predictions recorded before the corresponding results.  $\Delta$  is against Muon at the run’s learning rate (the early predictions predate the re-tuning) unless marked “vs tuned”.

|     | Prediction                                                                                                                            | Outcome                                                                                                                                                                    |
|-----|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P6  | Input-side SOAP basis keeps $\geq \frac{2}{3}$ of SOAP-Muon’s gain; standard basis $\leq \frac{1}{3}$                                 | Held: 0.78 (0.77–0.83 on 4 seeds); 0.18                                                                                                                                    |
| P7  | SOAP with the gradient’s instead of the momentum’s second moment differs by $\geq 0.005$                                              | Failed: 0.0003 apart                                                                                                                                                       |
| P8  | If the outlier sets Muon’s learning-rate ceiling, outlier-capped Muon at 0.02 beats Muon at 0.01 by $\geq 0.005$                      | Failed: +0.0093                                                                                                                                                            |
| P9  | Muon at 0.007 within $\pm 0.004$ of Muon at 0.01                                                                                      | Failed: $-0.0083$                                                                                                                                                          |
| P10 | Some data-norm run reaches $\Delta \leq -0.005$                                                                                       | Held: $\alpha = \frac{1}{4}$ gives $-0.0255$                                                                                                                               |
| P11 | Input-side SOAP replicates with $\Delta \leq -0.015$ on a second seed                                                                 | Held: $-0.0223$                                                                                                                                                            |
| P12 | Activation basis keeps $\geq \frac{2}{3}$ of input-side SOAP’s gain                                                                   | Held: 0.96                                                                                                                                                                 |
| P13 | One denominator per eigendirection keeps $\leq \frac{1}{3}$ of input-side SOAP’s gain                                                 | Failed: 0.68 (0.53 of SOAP-Muon)                                                                                                                                           |
| P14 | PD beats Muon, each at its best learning rate, by $\leq -0.010$ on every seed                                                         | Held: $-0.0172$ to $-0.0200$ (6 seeds)                                                                                                                                     |
| P15 | PD’s optimum $\alpha \in [\frac{1}{4}, \frac{3}{8}]$ , and $\alpha = \frac{1}{2}$ is worse than $\frac{1}{4}$                         | Held: optimum at $\frac{1}{4}$ ; $\frac{1}{2}$ worst                                                                                                                       |
| P16 | Magnitude-matched Muon keeps $< \frac{1}{3}$ of PD’s gain                                                                             | Failed literally: 0.39 (vs tuned $\approx 0.10$ )                                                                                                                          |
| P17 | $S+PD - PD \geq -0.005$                                                                                                               | Failed: $-0.0112$                                                                                                                                                          |
| P18 | PD’s gain persists at $2\times$ tokens: $\Delta \leq -0.010$ on both pairs (vs tuned)                                                 | Held: $-0.0164$ , $-0.0145$                                                                                                                                                |
| P19 | Output-side SOAP in data-norm coordinates keeps $\geq \frac{2}{3}$ of $S+PD - PD$                                                     | Held: 0.79                                                                                                                                                                 |
| P20 | Per-kind $\alpha = \gamma/2$ (finite-difference $\gamma$ ) improves on uniform $\frac{1}{4}$ by $\geq 0.002$ on both seeds (vs tuned) | Failed: $-0.0007$ , $+0.0019$                                                                                                                                              |
| P21 | Damped full data norm (damping 0.1) within 0.003 of PD or better (vs tuned)                                                           | Failed: $-0.0017$ , $-0.0066$ vs $-0.0172$ , $-0.0181$                                                                                                                     |
| P22 | At width 768, $S+PD$ beats tuned Muon by more than PD does, and beats S on both seeds (vs tuned)                                      | First part held ( $-0.0241$ , $-0.0270$ vs PD $-0.0118$ , $-0.0169$ ). Second part held on the one completed seed ( $-0.0241$ vs $-0.0183$ ); the other S seed was stopped |